# MAHARASHTRA STATE WATER TESTING CENTRAL LAB

ENVIRONMENTAL TESTING DIVISION * WATER QUALITY ANALYSIS & MONITORING SECTION

Testing Methodologies: APHA 23rd Edition / Bureau of Indian Standards (IS 3025 Baseline)

# WATER QUALITY ANALYSIS REPORT

TEST REPORT OF WATER SAMPLE ANALYSIS AND QUALITY EVALUATION

## 1. SAMPLE, LOCATION & TESTING PARTICULARS

| | | | |
|---|---|---|---|
| **Report Number:** | REP-20261105-D5BDF5 | **Sample Identifier:** | SMP-20261105233603 |
| **Water Body Name:** | Krishna River - Test Point | **Water Body Type:** | River |
| **Sample Location:** | Sangli Ghat, Maharashtra | **GPS Coordinates:** | 16.8524 N, 74.5815 E |
| **Sample Matrix / Type:** | surface_water | **Sample Collection Date:** | 05-Nov-2026 18:06 IST |
| **Testing / Analysis Date:** | 05-Nov-2026 18:06 IST | **Report Generation Date:** | 05-Nov-2026 18:06 IST |
| **Testing Protocol / Standard:** | Standard Methods (APHA 23rd Edition) | **Technician / Analyst:** | Dr. Tester V2 |

## 2. WATER QUALITY INDEX (WQI) & BASELINE ASSESSMENT

| | | | |
|---|---|---|---|
| **Predicted Water Quality Index (WQI):** | **41.83 / 100** | **Baseline Drinking Safety Screening:** | **SAFE** |
| **Quality Grading / Classification:** | Grade B - Good Quality (Suitable | **Regulatory Baseline Reference:** | CPCB Class C / BIS IS 10500:2012 |
| Analytical Assessment Model: | Decision Tree | Screening Scope: | Baseline Physical-Chemical-Biological |

## 3. MEASURED PARAMETERS & ANALYTICAL TEST RESULTS (14 PARAMETERS)

| No. | Tested Parameter | Reference Test Method | Unit | Standard Baseline Limit | Measured | Status |
|---|---|---|---|---|---|---|
| 1 | **pH** | APHA 4500-H+ B / IS 3025 (Pt | pH Units | 6.5 - 8.5 | **7.45** | Within |
| 2 | **Dissolved Oxygen (DO)** | APHA 4500-O C / IS 3025 (Pt | mg/L | >= 4.0 | **6.80** | Within |
| 3 | **Biochemical Oxygen Demand** | APHA 5210 B / IS 3025 (Pt 44 | mg/L | <= 3.0 | **2.10** | Within |
| 4 | **Chemical Oxygen Demand (CO** | APHA 5220 B / IS 3025 (Pt 58 | mg/L | <= 250.0 | **12.00** | Within |
| 5 | **Turbidity** | Nephelometric / IS 3025 (Pt | NTU | <= 5.0 | **3.80** | Within |
| 6 | **Total Dissolved Solids (TD** | Gravimetric / IS 3025 (Pt 16 | mg/L | <= 500.0 | **290.00** | Within |
| 7 | **Electrical Conductivity (E** | Conductivity Meter / IS 3025 | uS/cm | <= 1500.0 | **N/A** | N/A |
| 8 | **Water Temperature** | Thermometric / IS 3025 (Pt 9 | deg C | Ambient (15 - 35) | **24.50** | Within |
| 9 | **Nitrate (as NO3-)** | UV Spectrophotometric / IS 3 | mg/L | <= 45.0 | **4.20** | Within |
| 10 | **Phosphate (as PO4---)** | Stannous Chloride / APHA 450 | mg/L | <= 0.10 | **0.35** | Exceeds |
| 11 | **Chloride (as Cl-)** | Argentometric / IS 3025 (Pt | mg/L | <= 250.0 | **32.00** | Within |
| 12 | **Total Coliform** | Multiple Tube Fermentation / | MPN/100 mL | <= 5000 | **45.00** | Within |
| 13 | **Fecal Coliform** | Membrane Filtration / IS 162 | MPN/100 mL | <= 500 | **10.00** | Within |
| 14 | **Sulphate (as SO4--)** | Turbidimetric / IS 3025 (Pt | mg/L | <= 200.0 | **28.00** | Within |
| 15 | **Ammonia Nitrogen (NH3-N)** | Phenate Method / APHA 4500-N | mg/L | <= 0.50 | **0.25** | Within |

## 4. OBSERVATIONS & RECOMMENDATIONS

- Analytical parameters conform to standard baseline screening criteria (CPCB Designated Best Use Class C / IS 10500:2012).
- Recommended Treatment: Standard conventional filtration (rapid sand filtration) and controlled disinfection (chlorination /
- Surveillance: Maintain regular bi-weekly monitoring schedule to track seasonal hydrological and microbiological variations.

## 5. VERIFICATION & AUTHORIZATION

**Reporting Laboratory Analyst:**

**Dr. Tester V2**

Designation: Authorized Laboratory Analyst / Environmental Chemist

Analysis Date: 05-Nov-2026 18:06 IST

**Authorized Signatory Verification:**

*[ Signature of Authorized Signatory ]*

Quality Assurance Officer / Lab In-Charge

## 6. REPORT AUTHENTICITY DETAILS & REGULATORY DISCLAIMER

1. AUTHENTICITY NOTICE: This document is an electronically generated water quality analytical report compiled from laboratory measureme
2. STATUTORY DISTINCTION: This electronically generated report does NOT constitute an officially accredited statutory test certificate
3. SCOPE OF RESULTS: Analytical test results relate solely to the sample as tested and submitted for the specified monitoring point. Th